

Cypermethrin Insecticides Fail to Illicit Significant Epigenetic Changes in Acetylcholinesterase Gene Expression in *Blaberus Discoidalis*

Abstract:

Blaberus Discoidalis and other cockroaches are a common household pest in many parts of the world. Cockroaches are characterized by their resilience to lack of resources, extreme temperatures, and their ability to passively carry pathogenic bacteria. Meanwhile, Cypermethrin and other pyrethroid based insecticides have gained popularity in recent years due to their high potency for these insects and seemingly low mammalian toxicity. Pyrethroid insecticides are classified as neuro-toxins; however, the exact effect of these chemical compounds on gene expression in the species *Blaberus Discoidalis* remains to be elucidated. In this experiment, quantitative polymerase chain reaction techniques were utilized to explore the effect of Cypermethrin insecticides on arthropod gene expression. It was found that no significant change in acetylcholinesterase gene expression was observed; however, this may be due to a lack of statistical power obtained within the study. Overall, further study is needed to determine the exact mechanistic changes elicited by the use of pyrethroid insecticides on cockroaches in human homes.

Introduction:

Blaberus Discoidalis and other cockroaches are a common household pest in many parts of the world. Cockroaches are characterized by their resilience to a lack of resources and extreme

temperatures, often making the eradication of these insects an arduous task (Tanaka, 2007). Research has elucidated that cockroaches are known to release several proteins which activate immunoglobulins and subsequently cause allergic reactions in humans (Santos et. al., 1999). Additionally, prior research has elicited that cockroaches are capable of carrying numerous strains of potentially pathogenic bacteria, including those with antibiotic resistance; making the elimination of these pests a pertinent issue (Elgderi et. al., 2006). Meanwhile, pyrethroid based insecticides have gained popularity in recent years due to their high potency for arthropods and seemingly low mammalian toxicity (Sushila et. al. 2006, and Rahman et.al., 2006). Based off the chemical structure of native compounds isolated from plants of the pyrethrum genus; these neurotoxins work in target species by preventing the closure of voltage gated sodium channels and by affecting acetylcholinesterase gene expression (Jacques et.al., 2002). Moreover, pyrethroid compounds are usually broken apart by sunlight within a couple days, minimizing their deleterious effects on both the environment and ground water resources (Holmstead, 1978). However, the exact effects of the synthetic pyrethroid, Cypermethrin, on gene expression in the species *Blaberus Discoidalis* remains to be elucidated. In this experiment, the effect of Cypermethrin on Acetylcholinesterase gene expression relative to expression of the house keeping gene, β -actin, was explored. It was hypothesized that Cypermethrin would alter Acetylcholinesterase gene expression relative to controls. *Blaberus Discoidalis* cockroaches were exposed to Cypermethrin insecticide at half the LD_{50} for 12 days (LD_{50} obtained from Rahman et. al., 2006). Nervous systems were then harvested from the arthropods and Quantitative polymerase chain reaction techniques (qPCR) were utilized to explore the effect of this insecticide on acetylcholinesterase mRNA. Statistical software was then employed to assess the significance of changes in acetylcholinesterase gene expression relative to β -actin controls.

Materials and Methods:

Subjects: Sixteen HattianTropical cockroaches (species: *Blaberus Discoidalis*) were obtained from Backyard Brains in Ann Arbor, Michigan, numbered, and randomly assigned to either control or Cypermethrin treatment. Subjects were adult cockroaches, at least 8 months of age, mixed gender, and housed in 4.5cm of coconut coir soil, with HydroBalls (Zoo Med Laboratories, San Louis Obispo, CA) prior to experimentation.

Experimental Design: Control subjects were housed in cage dimensions (45.6 cm, 23.7cm, 20.6cm) with Eco Earth compressed coconut fiber substrate (Zoo Med Laboratories inc., San Louis Obispo, CA). The subjects were given 3lb. Moist Soil Granules and Super Grow Roach Food from American RHS for subsistence. The subjects were housed in this environment along with an egg carton for twelve days whereby their nervous systems were harvested and gene expression was measured. Cypermethrin treated subjects were also housed in identical housing conditions with identical food and water. Cypermethrin was applied to the coconut substrate used as soil within the cage, and dosage was determined according to previous studies LD_{50} 303 μ g/animal (Gammon D.W., 2012). Dosage was applied to the cage according to 85% the LD_{50} for each animal and a resultant .09225 mL of Cypermethrin were applied/cage (Calculation 1). Cypermethrin and control cages were then housed in a controlled environment for 12 days whereupon their nervous systems were harvested and gene expression was measured.

Cockroach Nervous System Dissection: Cockroaches were isolated from their cages and submerged in an ice bath for 6 minutes whereby their legs and antennae were clipped. The abdominal cavity was then opened and the abdominal ganglia (caudal-most section of ventral nerve cord) were extracted from the host. Ventral nerve tissue was then homogenized in a solution of guanidine thiocyanate and 1-thioglycerol to inactivate RNases and disrupt nucleoprotein complexes. Homogenate was then frozen at -70°C.

RNA Extraction: RNA was isolated from tissue sample according to ReliaPrep™ Tissue Miniprep instructions. Tissue sample was first homogenized for 10 seconds to ensure the mini columns

would not clog, and the resulting liquid was sheared and then centrifuged for 3 minutes (14,000 x g). Pellet was removed and supernatant was combined with 170µl of ReliaPrep™ Isopropanol. The resulting mixture was vortexed and transferred to a ReliaPrep™ mini column and centrifuged (12,000-14000 x g) for 1 minute. Liquid was removed, and 500 µl of RNA Wash Solution was added to the mini column. The mini column was centrifuged for 30 seconds (12,000-14,000 x g). To ensure that the RNA extracted from the subjects was not contaminated with DNA from the host, DNase I incubation mix was prepared according to ReliaPrep™ Instructions (432µl of Yellow Core Buffer, 54µl of MnCl₂, 54 µl DNase I, 540µl total, 30 µl for each sample of RNA.) and the resulting DNase I incubation mix was added to the extracted RNA mixture and allowed to incubate at 20°–25°C for 20 minutes. The mini column was then washed with ReliaPrep™ Column Wash Solution, and centrifuged for 30 seconds (12,000-14,000 x g). Wash Solutions were discarded and the ReliaPrep™ mini column was placed into a new collection tube. ReliaPrep™ Nuclease Free Water was added to the mini column (30 µl per sample, 18 samples 540 µl total) and the resulting mixture was centrifuged for 1 minute (12,000-14,000 x g). Quantity of RNA was measured using Nanodrop and determined to be 10.7 ng/µl RNA (1.5 Ratio). RNA extracts were stored at –70°C.

Reverse Transcription cDNA synthesis for qPCR: Oligonucleotide Primers were diluted 100 fold, (Calculation 2) and 1 µl of this resulting primer mixture was mixed with 1.88 µl of RNA extract and 2.12 µl of Nuclease Free water for a final volume of 5 µl. This was done for each sample of Extracted RNA. Resulting mixtures were incubated at 70°C for 5 minutes and then immediately chilled in ice water for 5 minutes. The samples were then microcentrifuged and stored on ice. Reverse Transcriptase mix was prepared using: 80 µl GoScript™ 5X Reaction Buffer, 40 µl MgCl₂ (at 1.5–5.0mM) , 20 µl PCR Nucleotide Mix (final concentration 0.5mM each dNTP), 10 µl RNasin® Ribonuclease Inhibitor, 10 µl GoScript™ Reverse Transcriptase, 130 µl Nuclease-Free Water for a total of 300 µl. 15 µl of Reverse Transcription Mix was added to each of the 5 µl RNA primer mixtures prepared above, and the resulting samples were then annealed at 25°C for 5 minutes, and then extended in a heat block at 42°C for 1 hour.

qPCR: Substrate for qPCR was prepared using 2 μ l of cDNA and 18 μ l of Master Mix for each sample. Master Mix for AChE qPCR was prepared using: 663.2 μ l of Bio-Rad™ Universal SYBR® Green Supermix, and this was added to the AChE Primers: (Forward: CGTTATCATTGTCTCCAT, Reverse: ATTGTAGTGCCATTAGTT, Spans 1007 to 1119; 119 bp) 37.79 μ l of Forward Primer (10 μ M), 37.79 μ l of Reverse Primer (10 μ M), and 517.263 of DI water. This yielded 1260 μ l of MasterMix which was prepared for the 18 samples. 20 μ l of substrate (18 μ l Master Mix and 2 μ l of cDNA) was loaded into the Bio-Rad CFX Connect Thermal Cycler three times for each sample, along with three positive controls containing RNA extract from last semester, two lanes containing the RNA primers, and one lane containing just master mix. (Diagram 1).

Master Mix for β -Actin qPCR was prepared the same way except the oligonucleotide primers were correspondent to the β -Actin mRNA (forward: TCGCACACAGTACCAATCTATGAA, Reverse: CAAGTCACGACCAGCCAGATC, Spans 361 to 438; 78 bp) and the substrate was prepared using just 1 μ l of cDNA and 19 μ l of Master Mix for each sample. Master Mix was prepared using 800 μ l of Bio-Rad SsoAdvanced™ Universal SYBR® Green Supermix (2X), 48 μ l of Forward Primer (10 μ M), 48 μ l of Reverse primer (10 μ M), 624 of DI water yielding 1520 μ l of MasterMix which was then prepared for the 18 samples. The Thermal Cycler temperatures were 95°C for 10 seconds, Annealing Temperature 65°C for 10 seconds, and Extension temperature 72°C for 30 seconds. Thermal Cycler was set to ~40 cycles for both the AChE and β -Actin.

Statistical Analysis: Melt Curves were obtained for the qPCR data to ascertain if the Extracted RNA was pure (Figure 1). Raw data for Quantification Cycle Values (C_q) was averaged for each sample for qPCR and plotted in Microsoft Excel according to what experimental conditions the cockroaches were subjected to (Control vs. Cypermethrin). The C_q for AChE was subtracted from the C_q for β –Actin (Table 1) and the resulting numbers were exponentiated to 2. This expression was plotted in a new table (Table 2), which was used to calculate Mean averages between gene expression differences as well as standard

Melt curves for both Acetylcholinesterase and β -Actin revealed that the qPCR reactions were specific and therefore produced only one product of cDNA (Figure 1). Analysis of qPCR data showed that on average, AChE was expressed 8.31 times more than β -Actin when *Blaberus discoidalis* subjects were exposed to Cypermethrin (n=6, SE=3.29) (Figure 2). Additionally, qPCR results showed that Acetylcholinesterase was expressed on average 13.21 times more than β -Actin in control subjects (n=8, SE=5.54) (Figure 2). Wells A10 and B11 were blank during analysis, so cockroach gene expression for subjects 10 and 11 were averaged out of two samples instead of three (both from the Cypermethrin treatment group), (Diagram 1). Additionally, data from Sample 9 (also Cypermethrin treatment) was not included in the mean average for the pesticide group because it was more than 2 standard deviations away from the mean average. Standard error was calculated for the differences between expression levels of control and Pesticide groups with Microsoft Excel, and found to be 5.54 and 3.29 for Control and Cypermethrin Groups, respectively (Table 2). A pairwise t-test was performed using Microsoft Excel on the data assuming unequal variances with 11 degrees of freedom, and a p-value of 46.24% was obtained with a two tail t-value of 2.20 (Table 3).

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D	13	14	15	16		+	RNAF	RNAR	MM			
E												

F										
G										

Thermal Cycler Layout: Samples were loaded into wells according to Methods for subsequent qPCR reactions. “+” corresponds to RNA isolated from experimental procedures conducted last semester. RNAF and RNAR represent wells containing just the forward and reverse primer oligonucleotides described in the methods section. MM corresponds to master mix, and was a control where just the master mix was run. Each number and the three rows below it correspond to a sample of cDNA extracted from a single cockroach specimen. For example, three samples of cDNA were extracted from Cockroach 1 and loaded into 1A, 1B, and 1C, respectively. Wells 5E, 5F, and 5G were left blank and are colored in black. Wells 9A, 9B, and 9C correspond to wells which were deemed outliers by statistical analysis. Wells A10 and B11 were found to be blank when qPCR was run.

Calculation 1:

$$\frac{1.4 \mu g}{\text{Cage}} \times \frac{1g}{100 \text{ mg}} \times \frac{100 \text{ g}}{26g} = .061 \frac{g}{\text{cage}} \times \frac{37.85 \text{ mL}}{30 \text{ mL} \times \text{oz}} \times 1.5 = .09225 \frac{\text{mLs}}{\text{cage}}$$

Amount of Cypermethrin added to Pesticide Group. 1.4 µg per cage corresponds to about half the LD₅₀ of 303 µg per animal. Cypermethrin was obtained at 26% and added into the calculation. Unit conversions for ounces to milliliters were also used to get measurements from the insecticide bottle into SI units. At the end, the ~57% was scaled up by 1.5 to ~85% LD₅₀ to ensure that the Cypermethrin would have an effect on gene expression in the target species. This was decided in light of news that last semester’s class not having a significant result using a conservative amount of pesticide relative to the LD₅₀ of their target species.

Table 1:

	A	B	C	D
Group	Cq Mean B-actin	Cq Mean AChE 1	Cq B-actin - Cq AChE1	Relative Gene Exp. $2^{(\text{Cq B-actin}-\text{Cq AChE1})}$
Control	26.37	35.42	-9.050	1.89
Pesticide	39.17	40.77	-1.600	329.88
Control	26.83	31.42	-4.590	41.52
Control	28.24	37.04	-8.800	2.24
Pesticide	26.40	32.20	-5.800	17.95
Control	29.33	36.75	-7.420	5.84
Pesticide	29.19	35.26	-6.070	14.88
Control	32.20	37.19	-4.990	31.47
Control	23.81	31.97	-8.160	3.50
Pesticide	25.90	37.60	-11.700	0.30

Pesticide	24.22	34.96	-10.740	0.58
Pesticide	24.89	33.58	-8.690	2.42
Control	26.05	31.80	-5.750	18.58
Control	23.56	34.20	-10.640	0.63
Pesticide	25.64	31.83	-6.190	13.70

Averages of Raw Data: Columns A and B represent averages of C_q Raw data for β -Actin and AChE, respectively, obtained from wells subjected to qPCR procedures. Column C represents the difference obtained by subtracting C_q B-Actin values from C_q AChE1 Values. Column D represents the Values from Column C exponentiated to the value of 2.

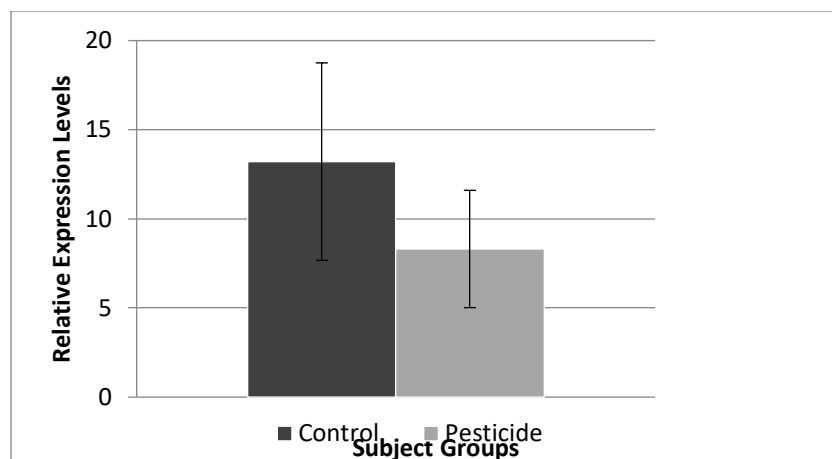
Table 2:

Group	Control	Pesticide
	1.89	
	41.52	17.95
	2.24	14.88
	5.84	0.30
	3.50	0.58
	31.47	2.42
	0.63	13.70
	18.58	
Mean	13.21	8.31
SEM	5.54	3.29

Gene Expression Values: Obtained from Column D of Table 1. A value on this table would represent the relative levels of Acetylcholinesterase compared to β -Actin. For example, a value of 2 would represent that there was twice as much acetylcholine gene expression in that subject as β -Actin gene expression. Mean values were taken by averaging the values above them and SEM corresponds to standard error calculations of the mean average values. Blank spaces on the table correspond to data that was either not obtained or excluded due to outlier status.

Table 3:

	<i>Variable 1</i>	<i>Variable 2</i>
Mean	13.20875	8.305
Variance	245.5801839	64.75031
Observations	8	6
Hypothesized Mean Difference	0	



Graphical Representation of Relative Gene Expression Levels: Bars show graphical quantification of average gene expression levels in controls (Dark) and Cypermethrin treatment. Error bars correspond to SEM measurements shown in Table 2. There is no significant difference between the two.

Discussion:

The graphical results in Figure 2 represent the main findings of this experiment; and that is that exposure to Cypermethrin for 12 days at 85% LD₅₀ caused a slight although insignificant decrease in acetylcholinesterase gene expression relative to controls in *Blaberus Discoidalis*. Indeed, while acetylcholinesterase gene expression was 13.21 times that of β -actin in controls, the prevalence of this gene expression fell to 8.31 times that of β -actin in Cypermethrin treatment groups. Prior research has elicited that Cypermethrin treatment has little effect on some housekeeping genes like β -actin (Shen et. al 2013). This in conjunction with the findings of our study suggest that perhaps there is a small change between Cypermethrin administration and acetylcholinesterase expression. The main drawback of this study; however, is that the p-value obtained by statistical analysis was 46.24%, which means that the odds are almost fifty percent that the findings of this study were due to chance. Likewise, the t-test statistic of 2.202 tells us that the findings of our study are more than two standard deviations away from the mean. This paired with the .05 α value (we were looking for a 95% certainty that our findings were not due

to chance) means that there is not sufficient evidence to reject the null hypothesis, and no assured conclusions can be made from this study.

Even though this study obtained an outrageously high p-value and subsequently could not find enough evidence to support a significant change in gene expression, the likely cause of this is the complications that arose when experimenters tested the Cypermethrin group. First off, one of the ventral nerve cords was lost during dissection, which reduced the n of the Cypermethrin group from 8 to 7 and subsequently made it more difficult to attain a significant result.

Furthermore, upon analysis of the data, another cockroach's results from the Cypermethrin group had to be repudiated from the results due to a change in gene expression that was over two standard deviations away from the mean of the group. While other subjects in the study were getting changes in gene expression with a magnitude between 2 and 15, this cockroach had a marked change in acetylcholinesterase gene expression of over 320, so in the interest of preventing a major distortion of this study's results, that cockroach's data had to be dismissed. Finally, two wells in the Cypermethrin group turned up blank when the thermal cycler raw data was obtained, so it is feasible that even the data that remained in the study could be skewed.

Since prior studies cite that Cypermethrin works both as a neurotoxin and a paralytic to target species, (Fournier et. al. 1994) and because acetylcholine is the neurotransmitter used in neuromuscular junctions, it is still plausible that changes in acetylcholinesterase gene expression are responsible for the behavioral effects that so many researchers observe. Since it is known that acetylcholinesterase is an enzyme that breaks down the excitatory neurotransmitter, acetylcholine, in neuromuscular junctions. One possible mechanism for this action could be that the retardation of acetylcholinesterase could work to flood neuromuscular junctions with acetylcholine, and effectively stall muscles in an activated state. Overall, it is pertinent that this

study be repeated to ascertain if there is really a significant change in acetylcholinesterase gene expression in the species, *Blaberus Discoidalis*, due to exposure to pyrethroid insecticides.